



From brain drain to brain circulation: the PSF supports national policy-makers in designing major R&I reforms

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PSF Insights

Having a competent workforce is a crucial asset for any nation's development and progress. Skilled professionals drive innovation, economic growth and social advancement. Factors such as better education, employment opportunities and higher salaries trigger individuals to emigrate. For instance, the movement of high-skilled workers has notably increased in Europe. In 2021, 32% of EU movers possessed high level of education, compared to 28% in 2016¹.

The circulation of talent between countries can yield beneficial outcomes. But Europe has also witnessed the negative effects of brain drain. Brain drain refers to the permanent emigration or outflow of skilled and talented individuals from their home regions to seek better opportunities and prospects elsewhere². Addressing this phenomenon is crucial, as it negatively impacts the sending regions, lowering their economic growth and generating significant regional differences within the EU. This article examines the challenges, recommendations and ongoing measures to reverse this trend, especially in eastern and southern European countries, bringing up specific examples from the PSF exercises.

Causes of brain drain: push and pull factors

Highly skilled workers may choose to emigrate for various reasons, including push factors in their home regions such as structural youth unemployment, poor education system or bad working conditions, and political instability. On the other hand, pull factors in the receiving region, such as higher living standards, access to healthcare, better job opportunities or cultural factors, may attract them³.

Economically, individuals are enticed by higher wages and improved job prospects abroad, while limited economic opportunities and lack of upward mobility at home discourage them from staying. For example, as per the Policy Support Facility (PSF) Specific Support Panel for Latvia, the country has suffered from a significant brain drain since the 1990s, primarily driven by wage disparities, creating a continuing trend⁴.

The quality of education and research facilities is another driver. Individuals leave their home countries because they have fewer chances for good education and limited access to well-known universities and research institutions, and they are attracted to the abundant academic resources in other countries. Romania is a prime example, as it is the European country most affected by talent emigration. PSF Country Romania highlights that the shortage of researchers is a major concern, with the lowest proportion among the EU27⁵.

¹ <https://ec.europa.eu/social/main.jsp?langId=en&catId=89&furtherNews=yes&newsId=10545>

² <https://cor.europa.eu/en/engage/studies/Documents/addressing-brain-drain/addressing-brain-drain.pdf>

³ <https://www.espon.eu/sites/default/files/attachments/3%20Emil%20Boc%20Brain%20Drain%20in%20the%20EU.pdf>

⁴ https://ec.europa.eu/research-and-innovation/sites/default/files/rio/report/55%2520Latvia_Final%2520Report.pdf

⁵ <https://op.europa.eu/en/publication-detail/-/publication/8a4a2522-eac3-11ec-a534-01aa75ed71a1/language-en/format-PDF/source-259353375>

Poor governance and corruption are another cluster of factors pushing individuals to emigrate from their home countries, as seen in the case of Greece. The reasons lie in economic challenges from the 2008 financial crisis, corruption concerns and lack of political trust. As highlighted in the PSF Country report, Greece's struggle to retain and attract highly skilled workers further demonstrates the impact of limited opportunities within the country⁶.

Common challenges and implications

The brain drain not only has economic consequences but also worsens regional disparities within the European Union. The countries facing significant emigration of skilled professionals tend to fall behind their European counterparts in terms of economic growth, innovation capacity and the country's level of development. This disparity perpetuates the cycle of brain drain as individuals continue to seek better opportunities in more developed regions. As a result, the existing regional imbalances within the EU widen further, having severe implications for their goal of catching up with other European countries⁷.

Another common challenge in affected countries is that mainly young, educated and skilled people are at the forefront of immigration, shrinking the working-age population and adding to the current problem of an ageing population. For instance, the PSF Country report on Montenegro highlights that the country is experiencing a loss of highly skilled workers to countries with larger and more developed labour markets, leading to a decline in the country's working-age population⁸. Similarly, Romania has faced a tough depopulation trend during the last decades, mainly due to emigration.

Zooming in on some critical sectors, brain drain exacerbates the shortage of skilled workers in them, especially within European countries facing challenges with their research and innovation (R&I) systems. This is, for example, the case of Croatia, where a significant proportion of emigrants possess tertiary education in medical, research, mechanical engineering and ICT fields⁹. This shortage has severe consequences, leading to challenges in providing essential services and reducing overall productivity in these sectors. Healthcare systems struggle to find qualified medical professionals, research institutions face difficulties in attracting and retaining talented researchers, and industries relying on technological expertise find it hard to stay competitive.

Moreover, brain drain has social implications, as it can perpetuate a brain drain mentality within society, where the emigration of talented professionals becomes normalised and perceived as the only viable option for personal and professional growth. This, in turn, discourages investment in home countries and limits overall development and progress¹⁰.

Strategies and recommendations to address brain drain

The joint effort between the European Union and European countries is crucial in addressing brain drain and its associated challenges. For that reason, the Commission has implemented the PSF instrument, which is a mechanism for offering support in designing, implementing and evaluating reforms for R&I systems in the EU Member States and countries associated with Horizon Europe. Through its different tools, the PSF looks into the challenges linked to brain drain at national level. Further, in the past years, the European Commission introduced several initiatives to address the impact of demographic changes¹¹, identifying a need to embed demographic considerations across EU policy and reduce disparities between regions. The 2020 Joint Action Plan between the European Committee of the Regions and the Commission aims to provide regions and cities with the latest data to mitigate brain drain and movement of human capital across EU countries and to address educational inequalities¹². Moreover, in 2023 the European Commission published a Communication on brain drain¹³ to investigate the drivers, consequences and recommendations for reversing it.

Additionally, as part of their national Recovery and Resilience Plans (RRPs), some Member States have implemented targeted reforms and investments to effectively counteract brain drain. For instance, Latvia has introduced a comprehensive higher education

⁶ <https://op.europa.eu/en/publication-detail/-/publication/fc6b0103-45ed-11ed-92ed-01aa75ed71a1/language-en/format-PDF/source-270143524>

⁷ <https://www.imf.org/en/Publications/Departmental-Papers-Policy-Papers/Issues/2019/07/11/Demographic-Headwinds-in-Central-and-Eastern-Europe-46992>

⁸ https://ec.europa.eu/research-and-innovation/sites/default/files/rio/report/SS%20Montenegro_Final%20Report.pdf

⁹ https://www.researchgate.net/publication/327836547_Written_by_European_Centre_of_Expertise_ECE_in_the_field_of_labour_law_employment_and_labour_market_policy_Labour_Market_Policy_Thematic_Review_2018_An_in-depth_analysis_of_the_emigration_of_skilled_l#pf6

¹⁰ <https://www.aeaweb.org/articles?id=10.1257/jel.50.3.681>

¹¹ <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52020DC0241&qid=1649670705351&from=EN>

¹² https://cor.europa.eu/en/engage/brochures/Documents/EU%20Annual%20Report%20on%20the%20State%20of%20Regions%20and%20Cities%202022/4739_Report%20State%20of%20R%20and%20C%202022_EN-N%20-%20main%20page.pdf

¹³ <https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13394-Brain-drain-mitigating-challenges-associated-with-population-decline-communication-en>

reform to enhance the quality and competitiveness of its institutions through structural changes in governance, funding, and human resources.¹⁴ Similarly, Slovenia has adopted a new Scientific Research and Innovation Activities Act and reformed its R&I governance, targeting brain drain by fostering research and innovation.¹⁵

At the country level, implementing multi-faceted initiatives is essential for tackling the problem. One key approach involves improving the attractiveness of the regions and enhancing domestic opportunities, creating favourable work environments, and offering competitive salaries and career prospects.¹⁶ For instance, in line with the PSF recommendations, Lithuania implemented strategies to enhance domestic opportunities by offering funds for ambitious research projects led by researchers working in foreign institutions. These schemes covered wages, research expenses and overhead costs, attracting top-notch scientists.

Similarly, initiatives to tackle the brain drain are in place to strengthen R&I ecosystems and contribute to scientific advancements and economic growth within the country. In line with the PSF recommendations, Romania has already announced a comprehensive plan to modernise and improve its R&I system. It includes initiatives to enhance merit-based funding, foster international partnerships, increase participation in Horizon Europe and promote technology transfer. As for Croatia, PSF recommendations are designed to support maximum impact from programmes foreseen in the 2021–2027 National Plan for Science, Technology and Innovation¹⁷, which aims to enhance researchers' conditions and attract public and private funding for R&I.

Furthermore, European countries can actively engage with their diaspora communities abroad. This can be achieved through specially designed initiatives like knowledge exchange programmes, mentorship schemes and entrepreneurship programmes. These initiatives aim to strengthen connections, promote collaboration and enable the diaspora to make meaningful contributions towards the development of their home countries. Moldova's government developed the 2015–2025 National Diaspora Strategy, encouraging skilled Moldovan migrants to return to the country and contribute to its development. Moldova's PSF recommendations align with these objectives, aiming to attract migrants back and encourage diaspora engagement in economic and social development.

Conclusion

Combating the brain drain is essential for promoting economic growth and innovation, particularly in regions with talent shortages. It requires a multi-faceted approach that focuses on enhancing domestic opportunities, investing in education and research, creating favourable work environments, and offering competitive salaries and career prospects. Furthermore, promoting institutional changes and inclusive working conditions within the labour market is crucial while encouraging common standards for European researchers and institutions. Both national governments and the European Union must collaborate in sustained efforts to effectively tackle the brain drain. In this regard, the implementation of PSF recommendations plays a crucial role. By aligning with the PSF and taking proactive measures, countries can mitigate the negative consequences and pave the way for ensuring that the best scientific talent contributes to a sustainable development.¹⁸

For further information:

[Visit the PSF website to learn more.](#)

The PSF's objective is to help Member States (MS)/Associated Countries to 'improve the design, implementation and evaluation of R&I policies'. The PSF provides expertise and practical support to MS in three major ways: PSF Country (formerly PSF Peer Reviews & Specific support to countries); PSF Challenge (including the PSF Mutual Learning Exercises, focused on specific and operational R&I challenges of interest in several volunteering countries); and PSF Open (which allows countries that have already benefitted from a PSF exercise to receive support to follow up on PSF recommendations). The PSF is funded under Horizon Europe.

¹⁴ https://commission.europa.eu/business-economy-euro/economic-recovery/recovery-and-resilience-facility/latvias-recovery-and-resilience-plan/latvias-recovery-and-resilience-supported-projects-reforms_en#reform-of-higher-education-and-scientific-excellence-and-governance

¹⁵ <https://www.gov.si/en/news/2023-03-25-national-assembly-adopts-an-amendment-to-the-scientific-research-and-innovation-activities-act/>

¹⁶ <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52019IR4645&rid=2>

¹⁷ https://ec.europa.eu/commission/presscorner/detail/en/ip_22_5065

¹⁸ https://diasporaforddevelopment.eu/wp-content/uploads/2020/04/CF_Moldova-v.5.pdf